

1 Strategic Imperative #2: Cloud adoption and Infrastructure Optimization; enabling Enterprise Digital Modernization

Our goal is to offer “**Full-lifecycle Managed Cloud Services**” to all customers and simplify and accelerate their cloud migration journey to enable innovation in a repeatable and reusable manner.

Specific services include:

- Build an Accelerator Platform for cloud adoption, comprising repeatable processes, architecture blueprints, turnkey services and landing zones with disaster recovery and failover
- Support Data Center Consolidation and IT Optimization through Private, Hybrid and Multi-Cloud implementations across our customer base
- Provide an Agile and DevSecOps platform with a service catalogue of templates to jump start and accelerate software development and systems modernization
- Infrastructure and Security as Code along with automated Security and Compliance (Authority To Operate -ATO)
- Provide a Single pane of glass for cloud administration, governance, billing and operational intelligence
- Intelligent automation with end to end continuous integration and continuous delivery (CI/CD) pipelines, including quality and security gates.
- Kubernetes based container orchestration with self-healing capabilities (DevSecOps / AIOps)

Investment #7: Turnkey Solutions for cloud adoption – Create a cloud environment to provide cloud adoption services such as landing zones, security as a code, CI/CD, ATO and container orchestration

1.1 Turnkey Solutions for 2020 Q1 Release 1.0 – March 30 (AWS & Google Cloud Focus)

Accelerator Platform Website & Sample Application

- 1) A responsive website to perform and integrate the seamless orchestration of turnkey solutions
- 2) Self-service catalog of templates to create Microservices and WebApps with baked-in design standards and security
- 3) Develop a sample web application (Movie Celebrities Information) that is modern and leverages containerized microservices and CI/CD orchestration

DevSecOps Orchestration into AWS and Google Cloud

- 4) End to end CI/CD pipeline orchestration with Quality and Security Gates for the sample application
- 5) Kubernetes based Container Orchestration for Zero Downtime Deployments (ZDD) in Openshift, GKE and AWS EKS and demonstrate elasticity and self-healing
 - a. For Release 1.0- Kubernetes environment will be pre-created in cloud platforms

“Cloud Infrastructure as Code and Security as Code”

- 6) Automated Standup and teardown of landing zones, i.e. environments in cloud using Terraform
 - a. AWS – Landing Zone along with EC2 and AWS EKS and deploy the sample app
- 7) Design for ATO Acceleration and Single Pane of Glass for billing and consumption across AWS and Azure
 - a. Aggregation of related ATO artifacts and map them to NIST 800-53 control families
 - b. Single Pane of Glass for billing and consumption across AWS and Azure

Cloud Adoption Playbook

- 8) Develop a Cloud Adoption playbook that comprises the conceptual and technical details and cost and benefits of cloud adoption journey for customers

1.2 Turnkey Solutions for 2020 Q2 Release 2 – April 1 to June 30 (Focus on Azure, GCP, ATO Acceleration and Single Pane of Glass)

- 1) Terraform Infrastructure as Code to create Landing Zones, i.e. environments in Azure and GCP Cloud
- 2) End to end CI/CD Pipeline orchestration into Azure with Quality and Security Gates using a sample application
- 3) Kubernetes based Container Orchestration for Zero Downtime Deployments (ZDD) in Azure and demonstrate elasticity and self-healing
- 4) ATO Acceleration and Single Pane of Glass for billing, reporting, operational intelligence across hybrid and multi-clouds using CloudChecker
- 5) Explore AIOps driven synthetic monitoring and telemetry over Kubernetes driven Intelligent Clusters to build proactive/predictive, self-healing and Intelligent Computing capabilities
- 6) Update Cloud Adoption playbook

2 Appendix: CONOPS and Architectural Patterns

Cloud Adoption Accelerator Platform Website



Provided by AceInfo Solutions, a Dovel company

Cloud Adoption and Digital Modernization Accelerator
 Empower Developers | Accelerate Innovation | Rapid Deployments

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Infrastructure as Code

- Create Cloud Infrastructure
- View Cloud Infrastructure
- Nexus/Docker Hub
- Terraform Templates
- Cloudformation Templates
- Ansible Templates
- Resources +

Application Development

- Create Microservice and Application
- Running Applications
- Github Repository
- Quality Assurance PAL
- Application Prototypes +
- Resources +

DevSecOps

- Code Security Scan
- View Code Scan Results
- Pipelines
- OpenShift
- API Management
- Regulatory Compliance
- Resources +

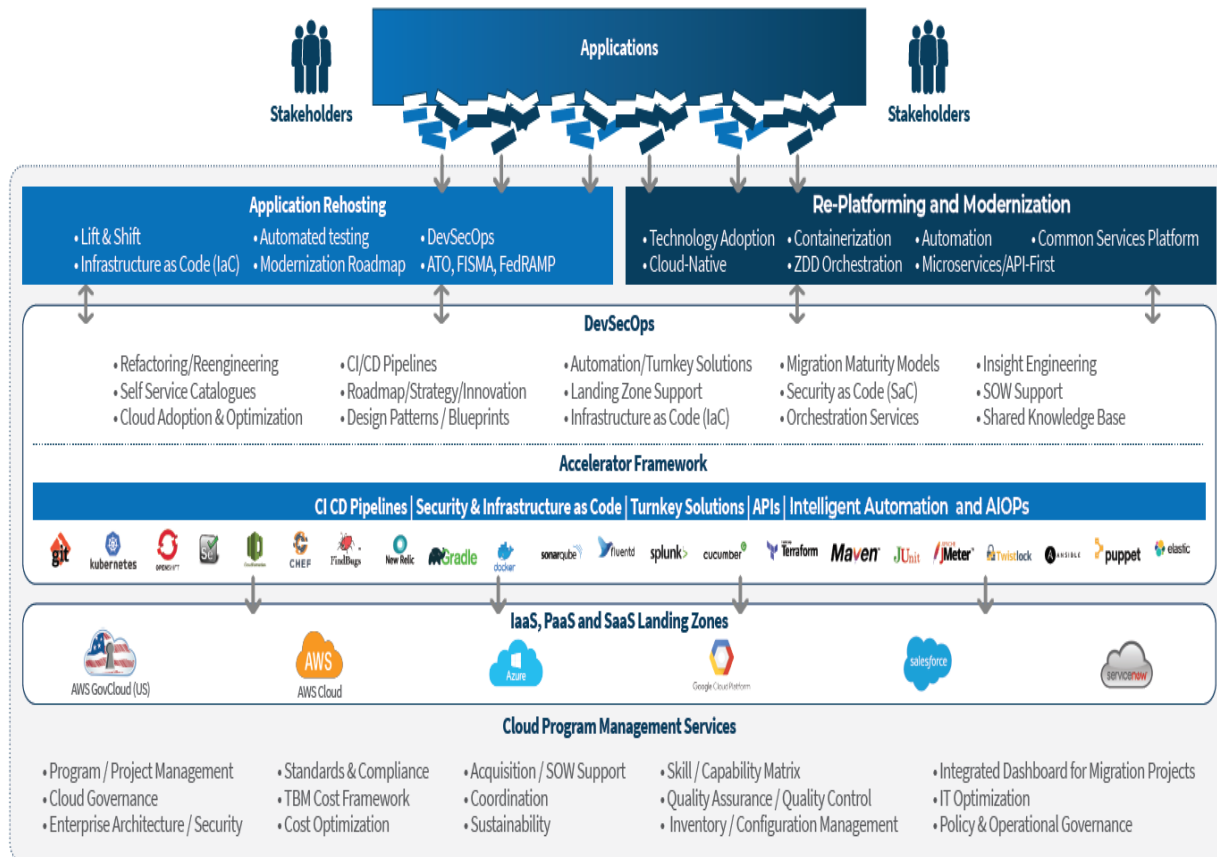
Monitoring and Logging

- Hygieia
- Splunk
- New Relic
- ELK Stack
- Slack
- Sonarqube
- Resources +

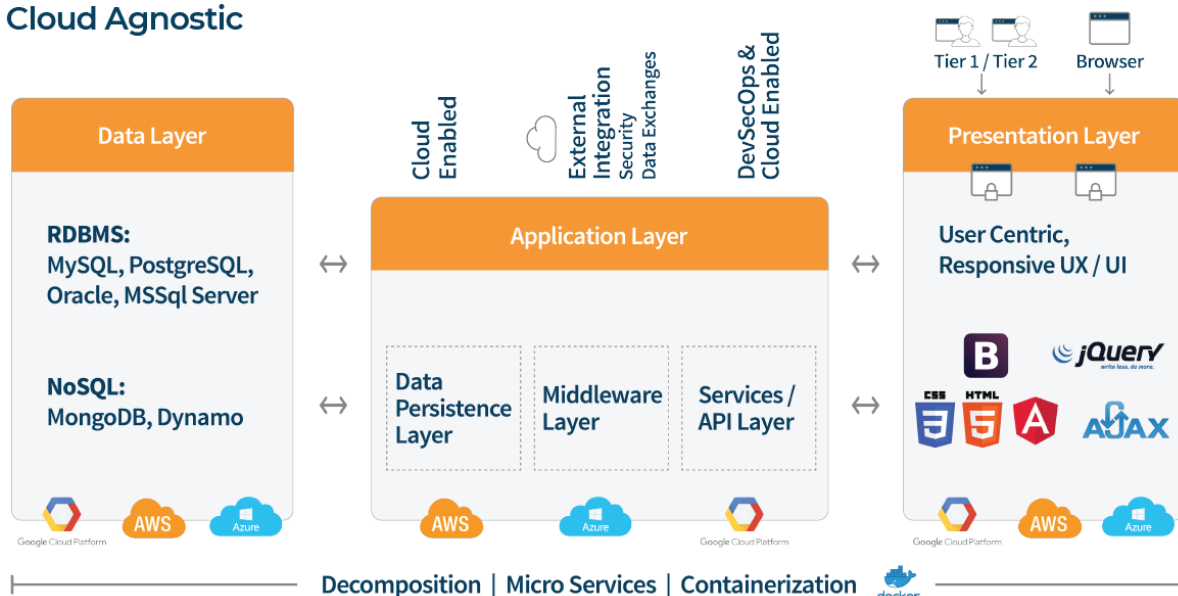
Intelligent Services / Analytics

- Text to Speech Conversion
- Machine Learning Models and APIs
- Facial Recognition & Media Analysis
- Social Media - Sentiment Analysis
- Resources +

Concept of Operations (CONOPS) for Cloud Adoption Accelerator Platform



Cloud Agnostic



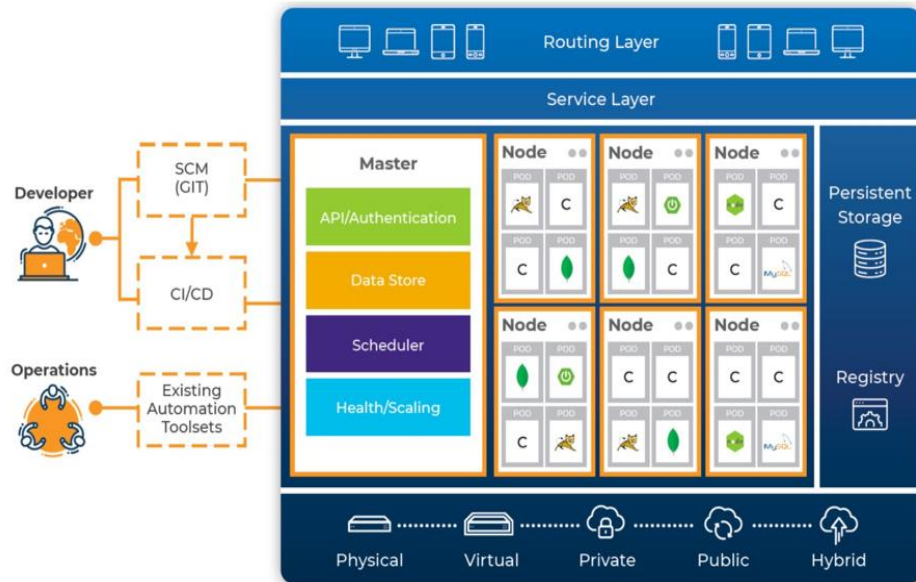
Kubernetes based High Availability and Self-Healing

■ Turnkey Solutions

- Cloud Adoption
- Immutability
- Security / ATO
- Scale and Failure
- AIOPs
- Self-Configure
- Self-Heal
- Self-Optimize
- Self-Protect

■ Turnkey Platforms

- Openshift
- Redhat Openshift
- Google Cloud (GKE)
- AWS (ECS, EKS)
- AZURE (CS, KS)



Insight Engineering and AIOPs for Predictive and Proactive Intelligence and Resilience

AIOPs

